1.3 Examining the rise of the agent era: Why agents?

AI agents and assistants have quickly moved from the main commodity in AI research to mainstream software development. An ever-growing list of tools and platforms assist in the construction and empowerment of agents. To an outsider, it may all seem like hype intended to inflate the value of some cool but overrated technology.

During the first few months after ChatGPT's initial release, a new discipline called *prompt engineering* was formed: users found that using various techniques and patterns in their prompts allowed them to generate better and more consistent output. However, users also realized that prompt engineering could only go so far.

Prompt engineering is still an excellent way to interact directly with LLMs such as ChatGPT. Over time, many users discovered that effective prompting required iteration, reflection, and more iteration. The first agent systems, such as AutoGPT, emerged from these discoveries, capturing the community's attention.

Figure 1.9 shows the original design of AutoGPT, one of the first autonomous agent systems. The agent is designed to iterate a planned sequence of tasks that it defines by looking at the user's goal. Through each task iteration of steps, the agent evaluates the goal and determines if the task is complete. If the task isn't complete, the agent may replan the steps and update the plan based on new knowledge or human feedback.